Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Your Wonderful Project Title

Topic outline

Author #1 Name (auth#1 NetID) - *auth#1 major*

Author #2 Name (auth#2 NetID) - *auth#2 major*

Supervision

- Supervisor #1 (sup#1-email-address)
- Supervisor #2 (sup#2-email-address)

Context

A very brief overview of the issues, and a short selection of related work. It's acceptable if the latter doesn't fit the topic perfectly; they are references that help circumscribe the perimeter of the problem you want to tackle.

For example, let's say you want to work on a secure online chat service. Your service aims to allow users to exchange messages without worrying about their privacy. You'll have to consider issues such as trust and malicious behaviours, both from users and from inconsiderate outsiders; [**?**] provides a service that contends with similar challenges. One of the hardest attacks to fend off, and that you'll probably have to consider is known as the *Sybil attack* [**?**], where users take on multiple identities within the service to raise their influence on the network. Opting for an architecture that has no central server offers two advantages: it removes the concern of a single point of attack, and it mitigates the power of nuisance of the service provider. But a decentralized architecture raises another tough challenge for messaging. In [**?**], Guerraoui discusses the complexity of delivering a sequence of messages in the same order to remote recipients.

Objectives

Start with a rough summary (one or two paragraphs) of the final result, in other words what you expect to achieve at the end of the semester. Let's go back to the secure online chat example. The summary should describe what features will be available to users; which of the issues mentioned in the context you'll focus your work on; if any, which off-the-shelf services and software you'll base your implementation on.

Finish with your work plan, with a set of steps you've laid out in order to reach your final result. For instance, in the case of the chat service example:

1. Design the user interface of the service, and use it to discuss the offered features and how they'll interact.
2. Come up with an architecture for an entirely serverless chat service.
3. Implement the service as a single local client application.
4. Deploy the service and test its operationality (loop back to the previous step if necessary).
5. Carry out a benchmark of attacks on the implementation to assess the security of the service.
6. Use the same benchmark on another service for comparison.